Circuit Board Step-by-Step Operations

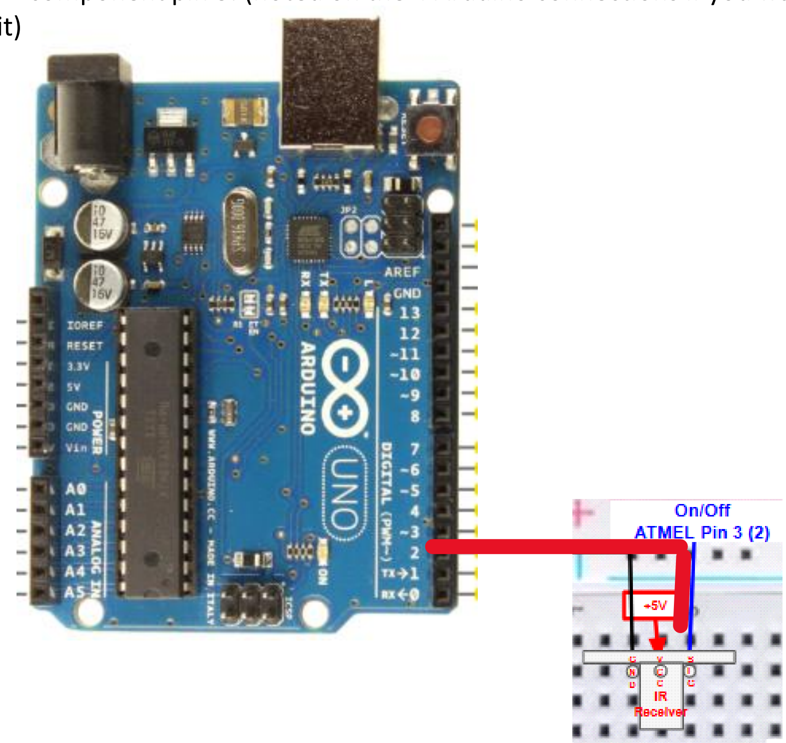
Tuesday, July 7, 2026 8:47 PM

Overview
The Remote Light Switch Circuit (RLSC) is used to allow multiple 12v led ceiling lights, wired in parallel, to be turned off/on with a handheld IR controller and/or a wall switch.

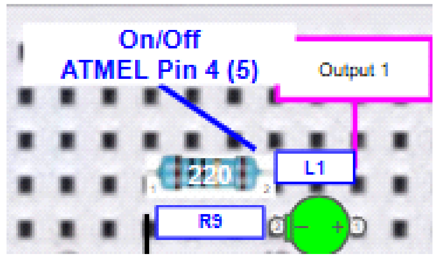
Components
- Arduino Uno
- KOOBOOK 2Pcs Digital 38KHz Infrared IR Receiver Sensor Module for Arduino
- (1) 47uF Radial Capacitor, 16V
- Nilight 50003R Automotive Set 5-Pin 30/40A 12V SPDT with Interlocking Relay Socket and Wiring Harness
- (11) 100K 1/2 watt resistors
- (2) 2N2222 npn transistors
- (1) BOJACK 1N4007 Rectifier Diode 1 A 1000 V DO-41 Axial 4007 1 amp 1000 Volt Electronic Silicon Diode
- (1) 3V 20ma 5mm Emitting Diode LED bulb
- (1) LM358P Dual Operational Amplifiers Op-Amp DIP8
- Any hand held IR Remote
- (3) KAKEMONO RV Recessed Ceiling Light Full Aluminum 12V LED Puck Light for Campervan Truck Caravan Boat Roof Under Cabinet Lighting Flush Mount, Daylight White 6000K

Arduino Sketch
The Arduino sketch excerpts below are for zone 1.
Additional code was created for 5 additional zones.

Operation
1) The wall light switch starts in the on position.
2) The handheld remote sends an IR signal to the IR Receiver to shut off the light.
3) The IR Receiver generates a pulse which is received by the Arduino Uno pin 2, which is connected to the onboard ATMEL component pin 3. (noted on the 4 Arduino connections if you wanted to create a stand alone ATMEL circuit)

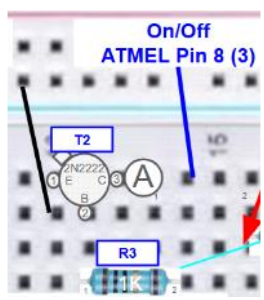


4) Arduino: `if(zone1_ledPinValue == HIGH && zone1_lightCircuitEnabled == true){ // on any ir received signal, reset the zoneX_lightCircuitEnabled variable if the wall switch was toggled off then on.`



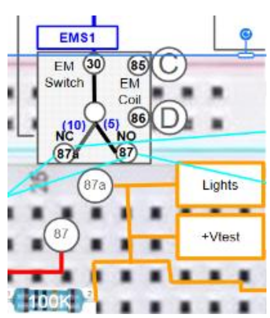
5) Arduino:
```
switch(IrReceiver.decodedIRData.decodedRawData){//determine which IR signal was received
      case button_1:
        lightCircuitEnabled = zone1_lightCircuitEnabled;
        button = button_1;
         //Serial.println("inside the switch: Button (1) pressed");
      break;
```

6) Arduino: `if(lightCircuitEnabled == false){//if light circuit is not enabled, enable it by raising the signal to T2. This turns off the light and puts the light circuit into a state waiting for the next signal`



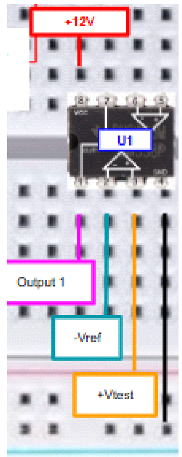
7) Arduino: `if(button == button_1){digitalWrite(zone1_t2Pin, HIGH);}`
Arduino: `Serial.println("15 second delay");` //wait 15 seconds for the Test Volage to drop below the Reference Voltage at the LM358 OpAmp
```
 delay(15000);
```

a) If the lights are on, the EM (electromechanical) switch is in the NO (Normally Open) position (wire 87), connecting the 12v power supply to the lights.



b) Raising the signal to T2 causes +Vtest to drop below -Vref

c) LM358 Output drops low and L1 LED shuts off

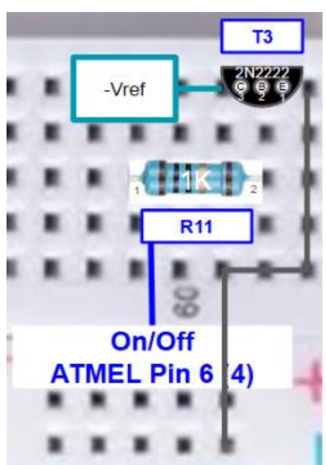


a) This also commands Vc on T3 to 12v, which in turn commands the EM switch to the NC (Normally Closed) position (wire 87A) and shuts off the lights.
b) +Vtest (~3volts) voltage is now applied to the light circuit.

9) Arduino: `if(button == button_1){digitalWrite(zone1_t2Pin, LOW);}//release the enablement signal to T2 now that the circuit is latched and the lights are off`

a) Debounce timer. Waiting 15 seconds allows +Vtest to drop low enough to create a latch, which will maintain the circuit state after the signal to T2 drops to low, and keep the EM switch in the NC position.

10) Arduino:
```
else{//if the lights were off, then raise T3 for 250ms to turn them on
      if(button == button_1){digitalWrite(zone1_t3Pin, HIGH);}
```



a) Sending another IR signal to the IR Receiver will raise a signal to T3 for 250ms and turn the lights back on.
b) The lightCircuitEnabled state will be reset

11) If the light switch is turned off the lightCircuitEnabled state will be reset and the RLSC will return to the original off state. The EM switch will return to the NO position, which is for the 12v power supply. Toggle the light switch to the on position to turn on the lights. This returns the system to step 4) above.